

LECTURE 7: CONVERGENCE OF RANDOM VARIABLES AND LAWS OF LARGE NUMBERS

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to elre@princeton.edu, thanks in advance!

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INTRODUCTION: LIMIT OF THE SUM OF INDEPENDENT RANDOM VARIABLES

Suppose that we observe a sequence of random variables $X_1, X_2, \dots$ that are independent and identically distributed (or i.i.d.) with mean $\mathbb{E}X_1 = \mu$. Note that for any finite n , $\mathbb{E}[\sum_{i=1}^n X_i] = n\mu$ by linearity of expectation. A natural question is the following:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i = ?$$

Naturally, we would expect the answer to be μ : a result of this form (for various notions of convergence, under different assumptions, etc.) is known as a **law of large numbers (LLN)**.

However, what does this exactly mean, the limit of functions (random variables)? Actually, there are several standard ways to define this limit, and the answer depends on the definition.

For example,

(Chebyshev bound). Assume further that the second moments are also finite ($\mathbb{E}|X_1|^2 < \infty$). Then Chebyshev's inequality and properties of the variance imply that for any $t > 0$,

$$\mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \geq t \right) \leq \frac{\sum_{i,j} \text{Cov}(X_i, X_j)}{t^2 n^2} = \frac{\text{Var}(X_1)}{t^2 n} \quad (1)$$

since the independent random variables are uncorrelated (i.e., pairwise covariances are zero). This way to describe convergence is called **convergence in probability**.

NOTIONS OF CONVERGENCE FOR RANDOM VARIABLES

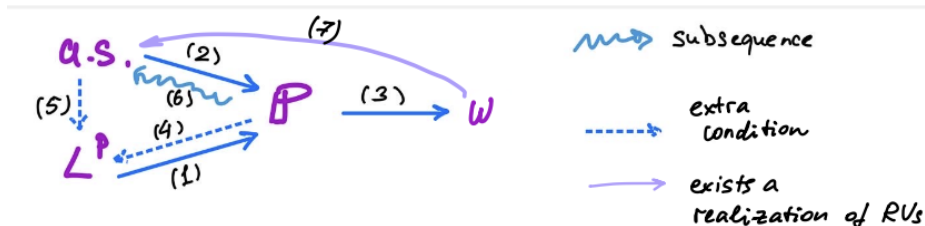
Other standard notions of convergence include

Definition 7.1. We say that a sequence of random variables $\{X_n\}_{n \geq 1}$ converges to a random variable X

- $X_n \xrightarrow{\mathbb{P}} X$ **in probability**, if $\mathbb{P}(|X_n - X| \geq \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$ for any $\varepsilon > 0$.
- $X_n \xrightarrow{\text{a.s.}} X$ **almost surely**, if $\mathbb{P}(\lim_{n \rightarrow \infty} X_n = X) = 1$.
- $X_n \xrightarrow{L^p} X$ **in L^p space**, if $\mathbb{E}(|X_n - X|^p) \rightarrow 0$ as $n \rightarrow \infty$.

- $X_n \xrightarrow{w} X$ **in distribution**, or **weakly**, if the corresponding distribution functions $F_{X_n}(t) \rightarrow F_X(t)$ as $n \rightarrow \infty$ for every $t \in \mathbb{R}$ where $F_X(t)$ is continuous.

Here is a scheme depicting some relations between these notions of convergence (solid arrow is an implication, dashed arrow is an implication under some additional conditions, curved arrow holds for a subsequence):



Proof of (1): Markov's inequality (simple exercise).

Proof of (2): Recall the formal definition of convergence for a sequence of numbers $X_n(\omega)$ (for a fixed ω): for any $\varepsilon > 0$, there exists n_0 so that for any $n \geq n_0$, $|X_n(\omega) - X(\omega)| \leq \varepsilon$ holds. Then we have the following:

$$\begin{aligned} 1 &= \mathbb{P}(\cup_{n_0=1}^{\infty} \cap_{n=n_0}^{\infty} (|X_n - X| \leq \varepsilon)) \\ &= \lim_{n_0 \rightarrow \infty} \mathbb{P}(\cap_{n=n_0}^{\infty} (|X_n - X| \leq \varepsilon)) \\ &\leq \lim_{n_0 \rightarrow \infty} \mathbb{P}(|X_{n_0} - X| \leq \varepsilon). \end{aligned}$$

Here, we used the continuity from below for property of probability measures (recall from Homework 1): if $B_1 \subseteq B_2 \subseteq \dots$ and $\cup_i B_i = B$, then $\mathbb{P}(B_n) \rightarrow \mathbb{P}(B)$ as $n \rightarrow \infty$. Here, $B_i = \cap_{n=i}^{\infty} (|X_n - X| \leq \varepsilon)$.

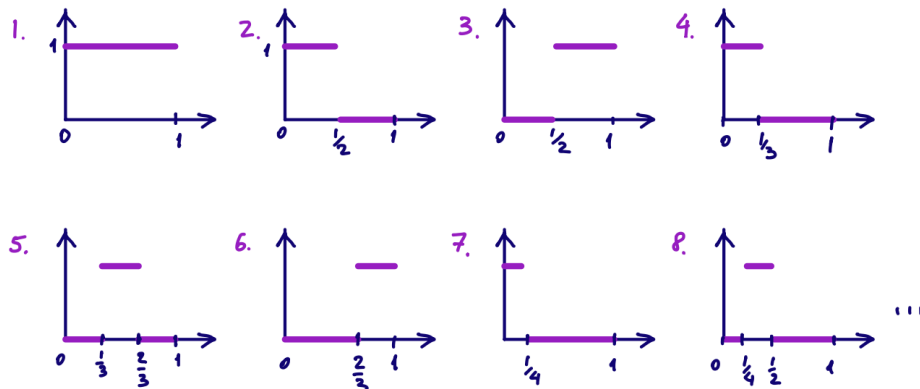
Proof of (3): Let t be a continuity point of F_X and take $\varepsilon > 0$. Then

$$F_{X_n}(t) = \mathbb{P}(X_n \leq t) \leq \mathbb{P}(X \leq t + \varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon).$$

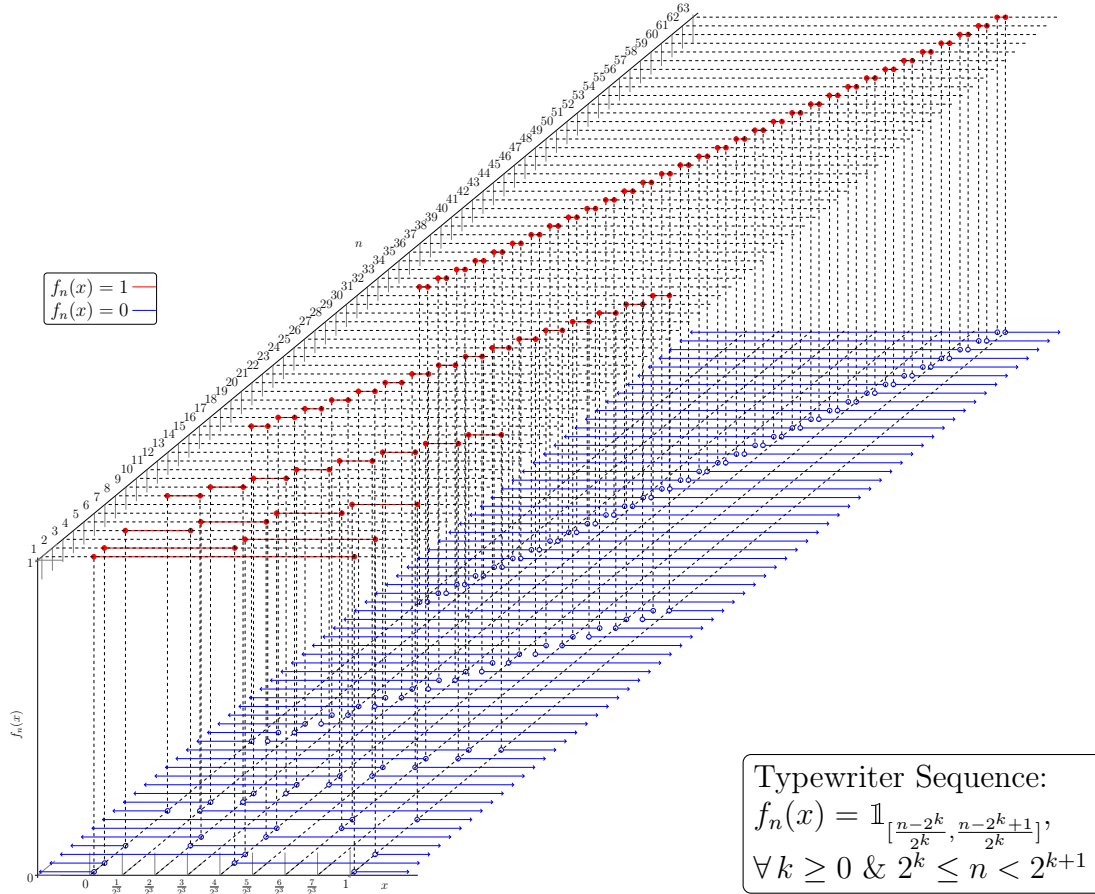
So, $\limsup F_{X_n}(t) \leq F_X(t + \varepsilon)$. Similarly, $\liminf F_{X_n}(t) \geq F_X(t - \varepsilon)$ for any $\varepsilon > 0$. Hence, $\lim F_{X_n}(t) = F_X(t)$.

Some counterexamples showing that some of the reverse implications do not hold:

- **Convergence in probability does not imply a.s. convergence:**



Below is the same example as above for a sequence that converges in probability but not almost surely, pretty picture due to the Stack Exchange:



We can see that the issue is the "moving mass" even though its support size tends to zero. Below, we show that it is always possible to select an a.s. convergent subsequence.

- **A.s. convergence does not imply L^p convergence:** consider

$$X_n = n^{1/p} \mathbb{1}_{[0, \frac{1}{\sqrt{n}}]}, \quad \text{then} \quad \mathbb{E}|X_n^p| = \int_0^{1/\sqrt{n}} n dx = \sqrt{n}.$$

Here, the issue is the unbounded nature of the random variable sequence. Even though the support of difference tends to zero, the values at this support tend to infinity. Next, we show that under additional assumptions on boundedness the reverse implication holds.

Implication (4):

Theorem 7.2. If $X_n \xrightarrow{\mathbb{P}} X$ and for some constant c , $|X_n| \leq c$ for all n , almost surely, then $X_n \xrightarrow{L^p} X$ for all $p \in [1, \infty)$.

Proof. First, note that $X_n - X$ is bounded almost surely. Indeed, $|X| - |X_n| \leq |X_n - X|$ by the triangle inequality, so for any fixed $\varepsilon > 0$, we have

$$\mathbb{P}(|X| \geq c + \varepsilon) \leq \mathbb{P}(|X| \geq |X_n| + \varepsilon) = \mathbb{P}(|X_n| - |X| \geq \varepsilon) \leq \mathbb{P}(|X_n - X| \geq \varepsilon) \rightarrow 0$$

as $n \rightarrow \infty$. Since the left hand side does not change with n , we deduce that $\mathbb{P}(|X| < c + \varepsilon) = 1$, and so, almost surely, for all n ,

$$|X - X_n| \leq |X| + |X_n| < 2c + \varepsilon.$$

This holds for any $\varepsilon > 0$, so $|X - X_n| \leq 2c$ a.s. Then,

$$\begin{aligned} \mathbb{E}|X_n - X|^p &= \mathbb{E}(|X_n - X|^p \mathbf{1}_{|X_n - X| \leq \varepsilon}) + \mathbb{E}(|X_n - X|^p \mathbf{1}_{|X_n - X| > \varepsilon}) \\ &\leq \varepsilon^p + (2c)^p \mathbb{P}(|X_n - X| > \varepsilon). \end{aligned}$$

Hence, from convergence in probability, $\limsup \mathbb{E}|X_n - X|^p \leq \varepsilon^p$ for any $\varepsilon > 0$ of our choice. We conclude that the limit must exist and equal zero. $\square$

Implication (5):

Definition 7.3 (Uniform integrability). We say that a sequence of random variables $\{X_n\}_{n \geq 1}$ is **uniformly integrable** if for any $\varepsilon > 0$, there exists $K > 0$ such that

$$\mathbb{E}(|X_n| \mathbf{1}_{\{|X_n| > K\}}) \leq \varepsilon \quad \text{for all } n.$$

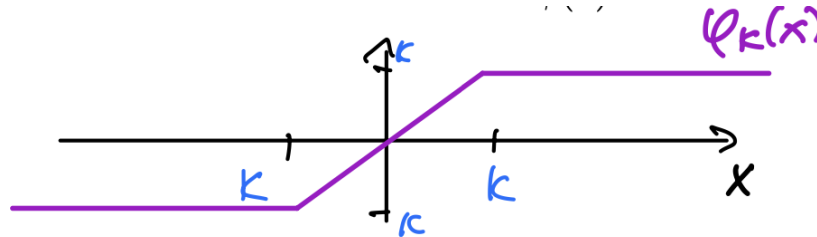
Exercise: check some sufficient conditions for uniform integrability:

- If $\sup_n \mathbb{E}(|X_n|^p)$ is finite for some $p > 1$, or
- If the sequence of random variables converges in L^1 ,

then $\{X_n\}_{n \geq 1}$ is uniformly integrable (see [Chatterjee, Propositions 8.3.2 and 8.3.5]).

Theorem 7.4. If $X_n \xrightarrow{\text{a.s.}} X$ and the sequence $\{X_n\}_{n \geq 1}$ is uniformly integrable, then $X_n \xrightarrow{L^1} X$ (and X is integrable).

Proof. Consider the following bounded continuous function $\phi(x) \equiv \phi_K(x)$ (where K is a constant that will be chosen to be large enough later):



Observe that for all n , we have the following inequalities:

$$\begin{aligned} \mathbb{E}|X_n - X| &\leq \mathbb{E}|X_n - \phi(X_n)| + \mathbb{E}|\phi(X_n) - \phi(X)| + \mathbb{E}|\phi(X) - X| \\ &\leq \mathbb{E}(|X_n| \mathbf{1}_{\{|X_n| > K\}}) + \mathbb{E}(|X| \mathbf{1}_{\{|X| > K\}}) + \mathbb{E}|\phi(X_n) - \phi(X)|. \end{aligned}$$

The last term tends to zero from almost sure convergence and the dominated convergence theorem (note that we needed truncation to apply the theorem!).

The first term is bounded by ε from uniform integrability (for any $\varepsilon > 0$, we can find K so that it is uniformly bounded).

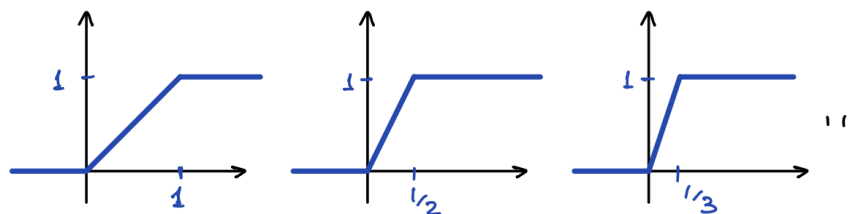
Finally, for the middle term, since

$$\mathbb{E}(|X_n|) \leq \mathbb{E}(|X_n| \mathbf{1}_{\{|X_n| \leq K\}}) + \mathbb{E}(|X_n| \mathbf{1}_{\{|X_n| > K\}}) \leq K + \varepsilon,$$

Fatou's lemma implies that X is integrable. Thus, by the dominated convergence theorem, $\mathbb{E}(|X| \mathbf{1}_{\{|X_n| > K\}}) \rightarrow 0$ as $K \rightarrow \infty$; in particular, this expectation is bounded by ε for K large enough. $\square$

CHARACTERIZATION OF WEAK CONVERGENCE AND IMPLICATION (7)

Why is weak convergence defined in terms of continuity points? Consider $\text{Unif}[0, \frac{1}{n}]$, the uniform distribution on $[0, \frac{1}{n}]$. What should the limit of this law be? What is the distribution function of an atom?



Observe that

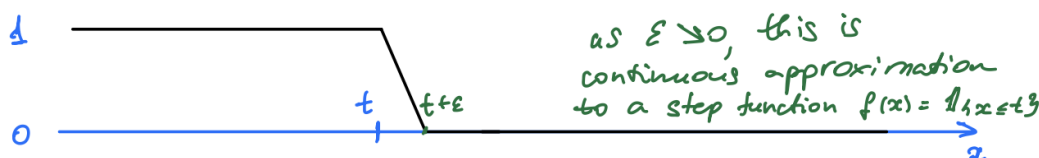
$F_{X_n}(t) \rightarrow F_X(t)$ if and only if $\mathbb{P}(X_n \leq t) \rightarrow \mathbb{P}(X \leq t)$ if and only if $\mathbb{E}[\mathbb{1}_{\{X_n \leq t\}}] \rightarrow \mathbb{E}[\mathbb{1}_{\{X \leq t\}}]$.

More generally,

Theorem 7.5. $F_{X_n}(t) \rightarrow F_X(t)$ for all points of continuity t of F_X if and only if for any continuous bounded function $g : \mathbb{R} \rightarrow \mathbb{R}$, we have $\mathbb{E}g(X_n) \rightarrow \mathbb{E}g(X)$.

Proof. Suppose that for any continuous bounded function $g : \mathbb{R} \rightarrow \mathbb{R}$, we have $\mathbb{E}g(X_n) \rightarrow \mathbb{E}g(X)$. Consider approximating the indicator random variable involved in the CDF by the following continuous (and bounded) functions:

$$g_{t,\varepsilon}(x) := \begin{cases} 1 & \text{if } x \leq t, \\ 1 - \frac{x-t}{\varepsilon} & \text{if } t < x < t + \varepsilon, \\ 0, & \text{if } x > t + \varepsilon. \end{cases}$$



Then for any random variable Y , we have the inequalities

$$\mathbb{1}_{\{Y \leq t\}}(y) \leq g_{t,\varepsilon}(y) \leq \mathbb{1}_{\{Y \leq t+\varepsilon\}}(y).$$

Thus, by the dominated convergence theorem,

$$\begin{aligned} \limsup_n \mathbb{P}(X_n \leq t) &= \limsup_n \mathbb{E}(\mathbb{1}_{\{X_n \leq t\}}) \leq \limsup_n \mathbb{E}g_{t,\varepsilon}(X_n) \\ &= \mathbb{E}g_{t,\varepsilon}(X) \leq \mathbb{E}\mathbb{1}_{\{X \leq t+\varepsilon\}} = \mathbb{P}(X \leq t + \varepsilon). \end{aligned}$$

That is, $\limsup_n F_{X_n}(t) \leq F_X(t)$ for all $t \in \mathbb{R}$. Similarly, we can bound $\liminf_n F_{X_n}$ from below (check this!) to get $\liminf_n F_{X_n}(t) \geq F_X(t-)$, recalling that $F_X(t-) := \mathbb{P}(X < t)$. If t is a point of continuity of F_X , then the upper and lower limits coincide, $F_X(t) = F_X(t-)$, and so $F_{X_n} \rightarrow F_X(t)$ as $n \rightarrow \infty$.

The converse result follows from the following:

Implication (7):

Proposition 7.6 (Version of Skorokhod's representation theorem). *Suppose that $X_n \xrightarrow{w} X$. Then there exist random variables $\{Y_n\}_{n=1}^\infty, Y$ on the probability space $([0, 1], \mathcal{B}([0, 1]), \text{Unif})$ such that $Y_n \stackrel{d}{=} X_n, Y \stackrel{d}{=} X$ and $Y_n \xrightarrow{\text{a.s.}} Y$.*

From Proposition 7.6, since $g(Y_n) \xrightarrow{\text{a.s.}} g(Y)$, we can use the dominated convergence theorem to take expectations on both sides and deduce that $\mathbb{E}g(X_n) = \mathbb{E}g(Y_n) \rightarrow \mathbb{E}g(Y) = \mathbb{E}g(X)$. $\square$

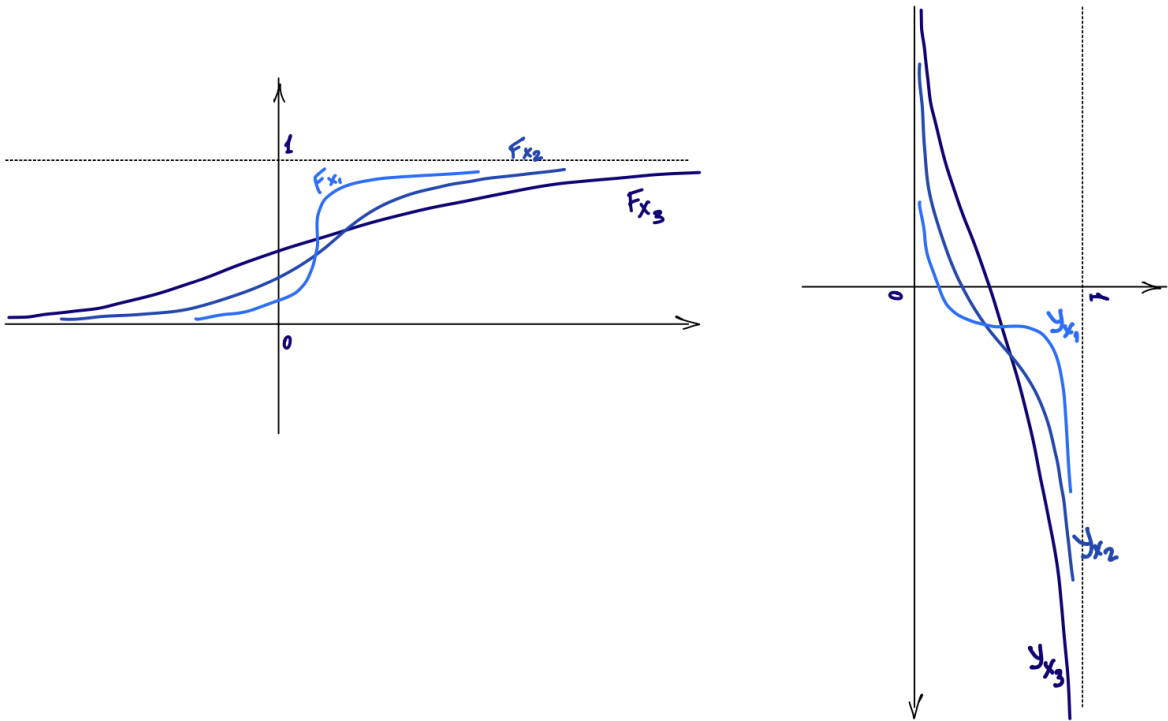
We saw that to prove the converse result in Theorem 7.5, it suffices to prove Proposition 7.6. We will only give the intuitive idea here (for the details, see, e.g., [Dembo, Theorem 3.2.7]).

Proof sketch for Proposition 7.6. Note that Proposition 7.6 implies that “just by rearranging” the realizations of random variables with the same distributions (as different functions realizing the same distribution), we can achieve a result stating that “the weakest convergence type implies the strongest”.

The idea behind the proof of the proposition is to look at the (generalized) inverses of the CDFs of X_n and X , denoting them by Y_n and Y :

$$Y_n(x) := \sup\{y \in \mathbb{R} : F_{X_n}(y) < x\}; \quad Y(x) := \sup\{y \in \mathbb{R} : F_X(y) < x\}.$$

Let us view Y_n and Y as almost everywhere continuous functions (or equivalently random variables) on $([0, 1], \mathcal{B}([0, 1]), \text{Unif})$ (recalling that the image of any CDF is in $[0, 1]$). One can show that $Y_n(x) \rightarrow Y(x)$ for all continuity points of Y_n, Y (this is where CDFs converge by weak convergence). But on the new space, this actually implies that $Y_n(x) \rightarrow Y(x)$ almost surely since the CDFs are continuous almost everywhere except at potentially countably many points of discontinuity.



$\square$

BOREL-CANTELLI LEMMA

Implication (6) requires an additional broadly applicable result, Borel-Cantelli lemma. Essentially, it outlines the conditions to convert a statement about probabilities/sizes of the events to an almost sure statement about the outcomes in Ω .

Theorem 7.7 (Borel-Cantelli lemma). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and $A_1, A_2, \dots$ be a sequence of events in $\mathcal{F}$.*

(a) *If $\sum_{i=1}^{\infty} \mathbb{P}(A_i) < \infty$, then*

$$\mathbb{P}(\{\omega : \omega \in A_n \text{ for infinitely many } n \in \mathbb{Z}_+\}) = 0.$$

Note that $\{\omega : \omega \in A_n \text{ for infinitely many } n \in \mathbb{Z}_+\} = \cap_{n=1}^{\infty} \cup_{m=n}^{\infty} A_m = \limsup_n A_n$.

(b) *If $\sum_{i=1}^{\infty} \mathbb{P}(A_i) = \infty$ and the events are mutually independent, then*

$$\mathbb{P}(\{\omega : \omega \in A_n \text{ for infinitely many } n \in \mathbb{Z}_+\}) = 1.$$

Proof. Part (a). Let $N(\omega) := \sum_i \mathbb{1}_{A_i}(\omega) \geq 0$. By MCT,

$$\mathbb{E}[N] = \sum_i \mathbb{P}(A_i) < \infty.$$

Thus, $\mathbb{P}(N = \infty) = 0$. Observe that this is exactly what we need: $N(\omega) = \infty$ if and only if $\omega \in A_n$ for infinitely many n .

Part (b). Note that from the representation of $\limsup_n A_n$, it is enough to show that $\mathbb{P}(\cup_{m=n}^{\infty} A_m) = 1$ for all $n \geq 1$. Consider the complement of this set:

$$\mathbb{P}\left(\bigcap_{m=n}^{\infty} A_m^c\right) = \lim_N \mathbb{P}\left(\bigcap_{m=n}^N A_m^c\right) = \lim_N \prod_{m=n}^N \mathbb{P}(A_m^c) = \lim_N \prod_{m=n}^N (1 - \mathbb{P}(A_m)),$$

where independence was (crucially) used in the second equality. Now, using the elementary inequality $1 - x \leq e^{-x}$ for any $x \geq 0$, we continue to estimate

$$\mathbb{P}\left(\bigcap_{m=n}^{\infty} A_m^c\right) \leq \lim_{N \rightarrow \infty} \exp\left(-\sum_{m=n}^N \mathbb{P}(A_m)\right) = \exp\left(-\sum_{m=n}^{\infty} \mathbb{P}(A_m)\right) = 0.$$

In the last step, we used the fact that $\sum_{m=n}^{\infty} \mathbb{P}(A_m) = \infty$ for any fixed n if the total sum is infinite (because $\sum_{m=1}^{n-1} \mathbb{P}(A_m) \leq n-1 < \infty$). $\square$

Implication (6):

Corollary 7.8. *If $X_n \xrightarrow{\mathbb{P}} X$, then there exists a subsequence $\{X_{n_k}\}_{k \geq 1}$ such that $X_{n_k} \xrightarrow{\text{a.s.}} X$.*

Proof. From convergence in probability, it follows that there exists a subsequence $\{X_{n_k}\}$ such that

$$\mathbb{P}\left(|X_{n_k} - X_{n_{k+1}}| > \frac{1}{2^k}\right) \leq \frac{1}{2^k}$$

for all k . By the Borel-Cantelli lemma (Theorem 7.7), since $\sum_{k=1}^{\infty} 2^{-k} < \infty$,

$$\mathbb{P}\left(|X_{n_k} - X_{n_{k+1}}| > \frac{1}{2^k} \text{ for infinitely many } n\right) = 0.$$

Thus, on a set of measure 1, the sequence $\{X_{n_k}(\omega)\}$ is convergent. Let us denote its limit by $Y(\omega)$ and (arbitrarily) define $Y(\omega)$ to be zero on the remaining set of measure zero. Then Y is the almost sure limit of $\{X_{n_k}\}$, and hence it is also the limit in probability. So,

$$\mathbb{P}(|X - Y| \geq \varepsilon) \leq \mathbb{P}(|X - X_{n_k}| \geq \varepsilon/2) + \mathbb{P}(|X_{n_k} - Y| \geq \varepsilon/2) \xrightarrow{k \rightarrow \infty} 0.$$

Since this holds for all $\varepsilon > 0$, we conclude that $X = Y$ almost surely. $\square$

Remark 7.9. It is easy to construct an example showing that independence is vital for part (b) of the Borel-Cantelli lemma; e.g., consider a sequence of random variables with $X_n \sim \text{Unif}[0, n^{-1}]$.

The Borel-Cantelli lemma is a result about the behavior of a chain of events “at infinity”; it can also be compared with the following:

Theorem 7.10 (Kolmogorov 0–1 law). *Let $\{X_n\}_{n=1}^\infty$ be a sequence of mutually independent random variables, and Υ be its tail σ -algebra defined as*

$$\Upsilon = \Upsilon(X_1, X_2, \dots) := \bigcap_{k=1}^\infty \sigma(X_k, X_{k+1}, \dots).$$

Then Υ is trivial; that is, for any $A \in \Upsilon$, either $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$.

Proof. Take any $A \in \Upsilon$. For any n , we have $A \in \sigma(X_{n+1}, X_{n+2}, \dots)$, which means that A is independent from $\sigma(X_1, \dots, X_n)$ (an exercise is to check that from mutual independence of random variables follows the independence with the whole σ -algebra). Consider

$$\mathcal{S} := \bigcup_{n=1}^\infty \sigma(X_1, \dots, X_n).$$

Check that (a) $\mathcal{S}$ is a σ -algebra; (b) A is independent from any set in it; and (c) $A \in \mathcal{S}$. Thus, A must be independent of itself, which implies that $\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)^2$, from which we conclude that $\mathbb{P}(A) \in \{0, 1\}$. $\square$

LAW OF LARGE NUMBERS

In the final section of this note, we are ready to put everything together to prove the Law of Large Numbers with respect to various notions of convergence.

Theorem 7.11 (Weak L^2 law of large numbers). *Let $X_1, X_2, \dots$ be a sequence of random variables with $\mathbb{E}X_i = \mu$ for all i . Assume that the sequence is uncorrelated ($\mathbb{E}[(X_i - \mu)(X_j - \mu)] = 0$ for all $i \neq j$) and has finite variance (for some constant $C > 0$, $\text{Var}(X_i) \leq C$ for all i). Then*

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{L^2} \mu.$$

Proof. From the properties of the variance and the assumptions, a straightforward calculation shows that

$$\mathbb{E} \left(\left| \frac{1}{n} \sum_i X_i - \mu \right|^2 \right) = \text{Var} \left(\frac{1}{n} \sum_i X_i \right) = \frac{Cn}{n^2} \rightarrow 0.$$

as $n \rightarrow \infty$. □

As discussed above, Theorem 7.11 implies convergence in probability (known as the *weak LLN*). However, the existence of second moments is not necessary for the weak LLN to hold, which can be proved by a sequence of truncations, similar to those in (2) below. The weak LLN also follows directly from the strong LLN, which we will prove in full generality (without any moment assumptions beyond the first moment):

Theorem 7.12 (Strong law of large numbers). *Let $X_1, X_2, \dots$ be a sequence of pairwise independent, identically distributed, and integrable random variables with $\mathbb{E}X_1 = \mu$. Then*

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} \mu.$$

Proof. We will prove the statement for non-negative random variables. An exercise is to check that decomposing an arbitrary random variable $X = X_+ - X_-$ into its positive and negative parts allows for the non-negativity assumption to be taken out. Moreover, it is actually enough to have only one of $\mathbb{E}X_+$ and $\mathbb{E}X_-$ finite.

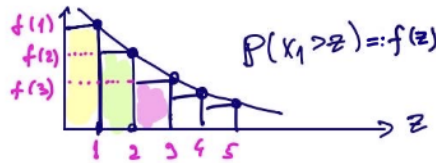
Step 1. First, we will define the convenient truncations

$$\bar{X}_i := X_i \mathbb{1}_{\{X_i \leq i\}}. \quad (2)$$

From the Borel-Cantelli lemma, it is enough to prove convergence of the random variables $\bar{X}_i$. Indeed,

$$\sum_{i=1}^{\infty} \mathbb{P}(X_i \neq \bar{X}_i) = \sum_{i=1}^{\infty} \mathbb{P}(X_i > i) = \sum_{i=1}^{\infty} \mathbb{P}(X_1 > i) \leq \int_0^{\infty} \mathbb{P}(X_1 > z) dz = \mathbb{E}X_1 < \infty.$$

Here, we estimated the integral of a non-increasing function by a sum (see the picture below), and used the integration by parts formula for the integral of a non-negative random variable (see Lecture 6).



$$f(1) \cdot 1 + f(2) \cdot 2 + \dots = \sum_{i \geq 1} f(i) \leq \int_0^{\infty} f(z) dz$$

Thus, by the Borel-Cantelli lemma (Theorem 7.7), $\mathbb{P}(X_i \neq \bar{X}_i \text{ for finitely many } i) = 1$. So,

$$\left| \frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \bar{X}_i \right| \xrightarrow{\text{a.s.}} 0,$$

which shows that it is enough to prove that $\frac{1}{n} \sum_{i=1}^n \bar{X}_i \xrightarrow{\text{a.s.}} \mathbb{E}X_1$.

Step 2. By the dominated convergence theorem, $\mathbb{E}(X_1 \mathbb{1}_{\{X_1 \leq i\}}) \rightarrow \mathbb{E}(X_1)$ as $i \rightarrow \infty$, using that X_1 is integrable. So, by the Cesàro mean property $a_n \rightarrow a$ implies $n^{-1} \sum_{i=1}^n a_i \rightarrow a$, we

have

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \bar{X}_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_1 \mathbb{1}_{\{X_1 \leq i\}}) \rightarrow \mathbb{E}(X_1).$$

Thus, it is enough to prove that $Z_n := \frac{1}{n} \sum_{i=1}^n (\bar{X}_i - \mathbb{E} \bar{X}_i) \xrightarrow{\text{a.s.}} 0$.

Step 3. The weak LLN (Theorem 7.11) implies that for any $\varepsilon > 0$,

$$\mathbb{P}(|Z_n| > \varepsilon) \leq \frac{1}{\varepsilon^2 n^2} \sum_{i=1}^n \text{Var}(\bar{X}_i).$$

This would also hold for any subsequence k_n instead of n . Now, the trick is to consider the subsequence $k_n := \lfloor \alpha^n \rfloor$ for some $\alpha > 1$. Our goal is to use the Borel-Cantelli lemma to go from convergence in probability for the full sequence to the almost sure convergence for such subsequences (compare with the proof of Corollary 7.8 above). In the last step, we show how varying α allows for the whole sequence to be sandwiched between such subsequences.

To apply Borel-Cantelli, we estimate the total probability of the “bad” events over the subsequence k_n :

$$\sum_{n=1}^{\infty} \mathbb{P}(|Z_{k_n}| > \varepsilon) \leq \sum_{n=1}^{\infty} \frac{1}{\varepsilon^2 k_n^2} \sum_{i=1}^{k_n} \text{Var}(\bar{X}_i) = \frac{1}{\varepsilon^2} \sum_{i=1}^{\infty} \text{Var}(\bar{X}_i) \sum_{n: k_n \geq i} \frac{1}{k_n^2}.$$

Now, the choice of k_n ensures the following useful property: there exists a constant C such that for any i ,

$$\sum_{n: k_n \geq i} \frac{1}{k_n^2} \leq \frac{C}{i^2}. \quad (3)$$

We leave verifying this as an exercise: the idea is that the sequence $\{k_n\}$ grows fast enough so that its tails (for $k_n \geq i$) decay in a summable way as $O(i^{-2})$. Continuing, using the fact that $\text{Var}(Y) \leq \mathbb{E}(Y^2)$ for any random variable Y and the monotone convergence theorem,

$$\sum_{n=1}^{\infty} \mathbb{P}(|Z_{k_n}| > \varepsilon) \leq \frac{C}{\varepsilon^2} \sum_{i=1}^{\infty} \frac{\mathbb{E}(\bar{X}_i^2)}{i^2} = \frac{C}{\varepsilon^2} \sum_{i=1}^{\infty} \frac{\mathbb{E}(X_1^2 \mathbb{1}_{\{X_1 \leq i\}})}{i^2} = \frac{C}{\varepsilon^2} \mathbb{E} \left(X_1^2 \sum_{i=1}^{\infty} \frac{\mathbb{1}_{\{X_1 \leq i\}}}{i^2} \right).$$

For the last term, note that

$$\sum_{i=1}^{\infty} \frac{\mathbb{1}_{\{X_1 \leq i\}}}{i^2} = \sum_{i \geq X_1} \frac{1}{i^2} \leq \frac{C'}{X_1}$$

for some constant C' . Thus, we conclude that $\sum_{n=1}^{\infty} \mathbb{P}(|Z_{k_n}| > \varepsilon) \leq \frac{C''}{\varepsilon^2} \mathbb{E}(X_1) < \infty$. Applying the Borel-Cantelli lemma, we deduce that Z_{k_n} converges to 0 almost surely for any $k_n = \lfloor \alpha^n \rfloor$ with $\alpha > 1$. (Actually, we proved that this holds for any increasing sequence of indices such that the property 3 holds.)

Step 4. Finally, note that for every n we can find an element k_ℓ of the subsequence such that $k_\ell \leq n < k_{\ell+1}$, and so

$$\frac{k_\ell}{k_{\ell+1}} \frac{\sum_{i=1}^{k_\ell} \bar{X}_i}{k_\ell} = \frac{\sum_{i=1}^{k_\ell} \bar{X}_i}{k_{\ell+1}} \leq \frac{\sum_{i=1}^n \bar{X}_i}{n} \leq \frac{\sum_{i=1}^{k_{\ell+1}} \bar{X}_i}{k_\ell} = \frac{k_{\ell+1}}{k_\ell} \frac{\sum_{i=1}^{k_{\ell+1}} \bar{X}_i}{k_{\ell+1}}.$$

This implies that for any $\alpha > 1$, as $n \rightarrow \infty$ (and so $\ell = \ell(n) \rightarrow \infty$), we have

$$\alpha^{-1} \mathbb{E}X_1 \leq \liminf_n \frac{\sum_{i=1}^n \bar{X}_i}{n} \leq \limsup_n \frac{\sum_{i=1}^n \bar{X}_i}{n} \leq \alpha \mathbb{E}X_1.$$

Hence, by taking $\alpha \rightarrow 1$, we obtain the desired result. $\square$

To recap the general process of the proof of the Strong LLN:

- We first truncate to be able to use convergence in probability (Chebyshev inequality).
- It gives us the bound on the probabilities of tail events $\{X_n > K\}$ of order n^{-1} , which is not summable to apply Borel-Cantelli lemma and claim a.s. convergence.
- So, we consider a *lacunary* sequence indexed by $n_k \sim \alpha^k$ which converges fast enough to apply Borel-Cantelli and converges a.s.
- All other subsequences have to converge to the same limit from the sandwiching property (Step 4), this is because our subsequence is not too lacunary as we send $\alpha \rightarrow 1$.